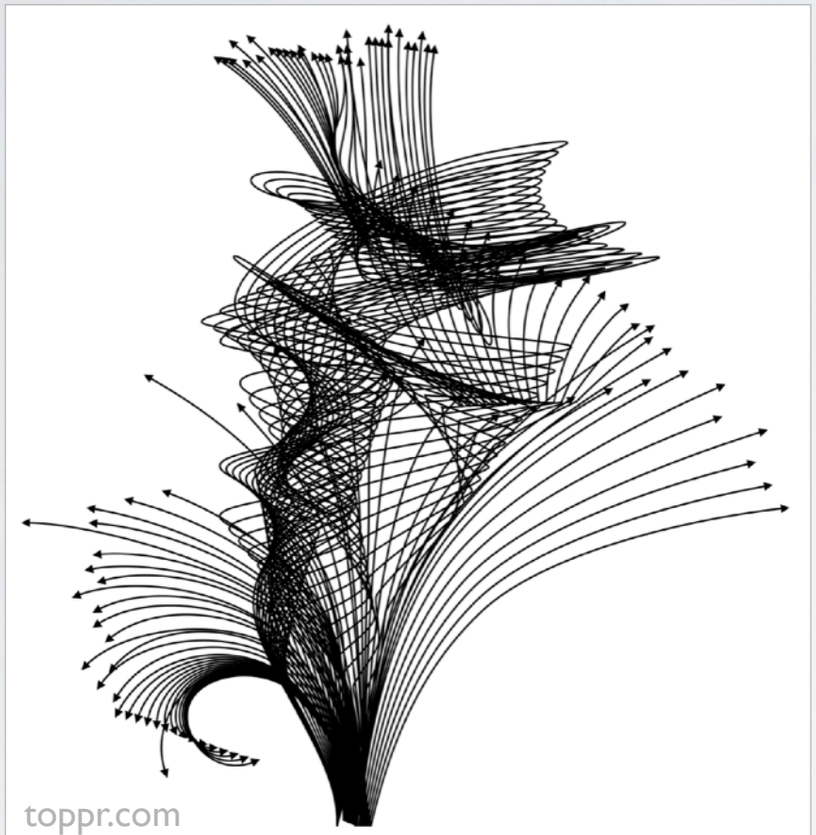
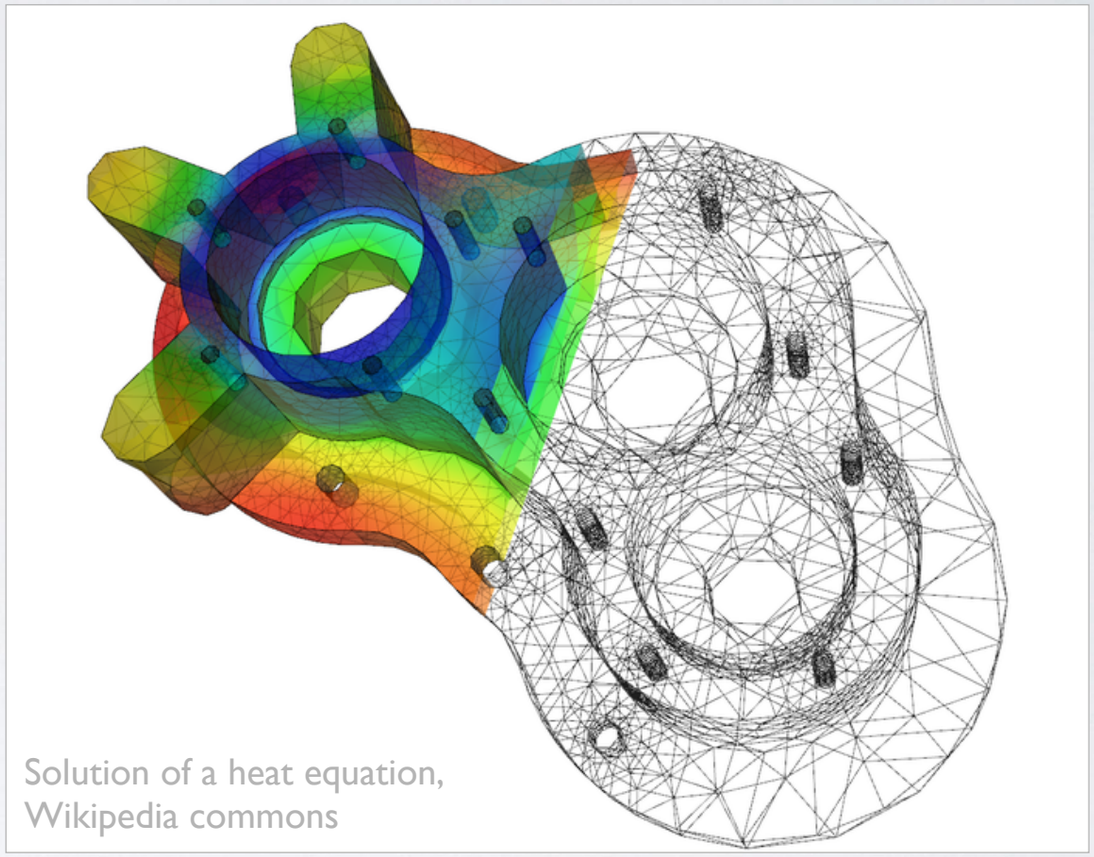
# Week 10 - #2

# Ordinary Differential Equations (II)



toppr.com



Solution of a heat equation,
Wikipedia commons

Today: Ch 8

Next Class: Ch 8

Ji-hoon Kim (Seoul National University)

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Despite his best efforts, the author apologizes for any inaccurate or outdated information and references.

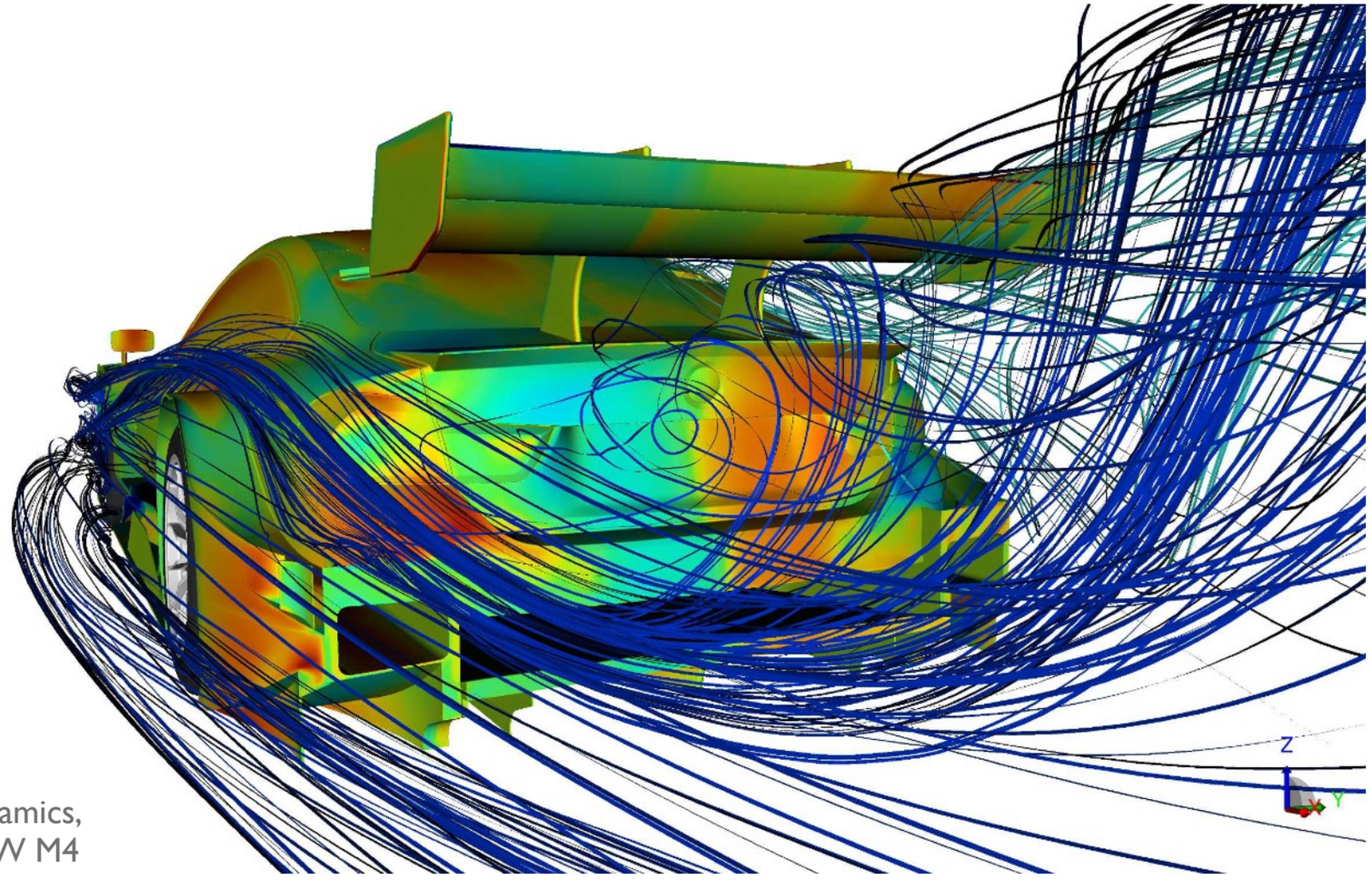
# Rudimentary Mathematical Methods of Physics (Fall 2026): Quiz #15

— [ open book and open note, but no cellphone or laptop, drop it off as you leave the class ] —

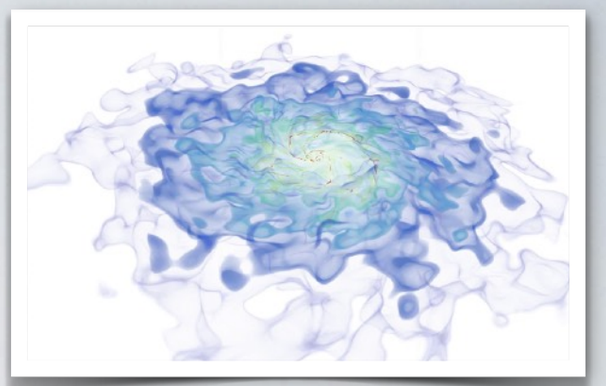
Please write down your name and student ID in the top right corner. (0.0 pt: no paper found with your name / 0.5 pt: paper found with your name and some answers / 1.0 pt: good answers)

1. Boas, Chapter 8, Section 1, Problem 2

2. Boas, Chapter 8, Section 2, Problem 4 (after reviewing Eqs.(2.1)-(2.3) on separable equations)




Computational fluid dynamics,
BMW M4



# Solving DE on WolframAlpha

# Solving DE (Section 8.3, Example 1)

WolframAlpha computational intelligence

solve (1+x^2)y'+6xy=2x

NATURAL LANGUAGE | MATH INPUT | EXTENDED KEYBOARD | EXAMPLES | UPLOAD | RANDOM

Input interpretation

solve $(1+x^2)\,y'(x)+6\,x\,y(x)=2\,x$

Result

Step-by-step solution

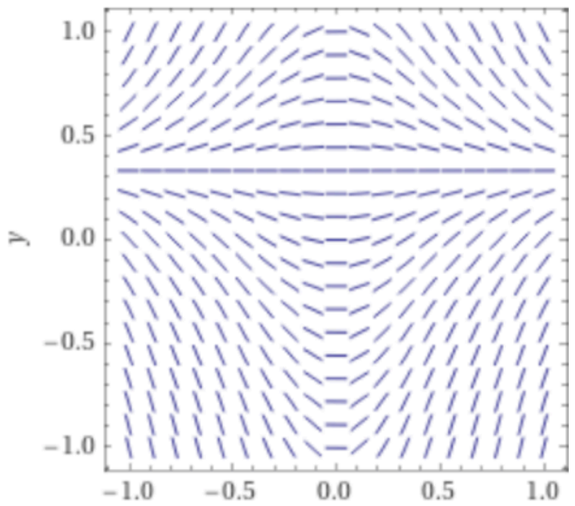
$$y(x)=\frac{c_1}{(x^2+1)^3}+\frac{x^2}{(x^2+1)^3}+\frac{x^6}{3\,(x^2+1)^3}+\frac{x^4}{(x^2+1)^3}$$

Separable equation

$$\frac{y'(x)}{2-6\,y(x)}=\frac{x}{1+x^2}$$

Slope field

Fewer points | More points | Slope field



# Solving DE (Section 8.4, Problem 7)

WolframAlpha computational intelligence

solve x^2 dy + (y^2-xy)dx = 0

NATURAL LANGUAGE | MATH INPUT | EXTENDED KEYBOARD | EXAMPLES | UPLOAD | RANDOM

Input

$$x^2\,dy + (y^2 - x\,y)\,dx = 0$$

ODE names

Bernoulli's equation

$$y'(x) = \frac{y(x)}{x} - \frac{y(x)^2}{x^2}$$

Bernoulli's equation »

Homogeneous equation

$$y'(x) = -\frac{\left(-1 + \frac{y(x)}{x}\right) y(x)}{x}$$

Homogeneous equation »

ODE classification

first-order nonlinear ordinary differential equation

Differential equation solution — Step-by-step solution

$$y(x) = \frac{x}{c_1 + \log(x)}$$